School of Future Tech

Case Study Report

on

Flappy Bird Clone with Scoreboard

by

Prince Rajesh Yadav
150096725032

Index

1. Introduction to the Case Study.
2. Problem Statement / Case Background (Abstract).
3. Problem Statement / Case Study Design.
4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study.
5. Problem Statement / Case Study Implementation Details and Snapshots.
6. Problem Statement / Case Study Results and Conclusion.
7. References

1. Introduction to the Case Study

Browser-based arcade games are popular for entertainment as well as learning the basics of game development. One of the most popular games is the classic Flappy Bird game, in which the user has to control a bird passing through obstacles. However, most browser-based games are simple and lack features such as scoring, competition, and user interaction.

In this case study, the focus is on designing and developing a Flappy Bird Clone with Leaderboard and Forum System. This is a web-based game in which the user can play by controlling a bird passing through obstacles. The user will be awarded points based on the successful passage of the bird through the obstacles. This game also includes a Leaderboard in which the user is ranked based on their highest score. Additionally, there is a Forum in which the user can post their comments about the game.

2. Problem Statement / Case Background (Abstract)

Background

Simple browser-based games are only concerned with the gaming aspect and do not have other elements such as score comparison and user interaction. When the leaderboard feature is not implemented, users cannot compare their scores with those of other users. Also, users cannot interact with each other through the communication feature.

For better user interaction, new elements have been added to the browser-based games.

Abstract

The case study discusses the design and development of a Flappy Bird Clone with Leaderboard and Forum Features using HTML, CSS, and JavaScript. The browser-based game has been implemented with gravity-based movement for the birds and random movement for the pipes, providing the feature of an endless runner game.

The score for the users is calculated and updated whenever the user passes the obstacles. The score and username of the users are stored locally, and the leaderboard is implemented to show the ranking of the users based on their maximum scores. The leaderboard does not store duplicate usernames and scores, and the score of the users is updated whenever they achieve a higher score than their previous maximum score. The forum page has been implemented to allow users to interact with each other by posting their comments.

3. Case Study Design

The Flappy Bird Clone system is designed as a simple modular structure consisting of multiple pages connected through navigation.

System Flow

Home Page → Play Game → Game Over → Enter Name → Leaderboard → Forum

Key Components

1. **Home Page**
    - Provides the main interface and navigation to other sections.
    - Includes links to Play, Leaderboard, and Forum pages.
2. **Game Page**
    - Displays the Flappy Bird gameplay area.
    - Implements bird movement, pipe generation, and collision detection.
    - Calculates score dynamically during gameplay.
    - Supports multiple control methods such as keyboard, mouse click, and touch input for better accessibility across devices.
3. **Game Over System**
    - Detects collisions with pipes or ground.
    - Displays final score.
    - Prompts the player to enter their name.
4. **Leaderboard System**
    - Stores player names and scores.
    - Displays rankings based on highest scores.
5. **Forum Page**

- Allows users to post comments about the game.
- Stores comments using browser storage.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key Methods and Techniques

1. Game Logic Implementation

- Gravity-based bird movement.
- Pipe generation at fixed time intervals.
- Collision detection between bird and obstacles.
- Score calculation when the bird passes pipes.
- Multiple input controls including keyboard (spacebar), mouse click, and touch/trackpad input for bird movement.

2. Data Storage

- Player scores stored using **localStorage**.
- Forum comments also stored using **localStorage**.

3. Leaderboard Algorithm

- Retrieve stored scores.
- Sort scores in descending order.
- Display player rankings.
- Update existing player scores instead of creating duplicate entries.

4. UI/UX Design

- Animated clouds and backgrounds.

- Responsive layout using CSS media queries.
- Tabular leaderboard structure with rank highlighting.
- • Responsive input controls supporting keyboard, mouse, and touch interactions.

Technology Stack Used for the Case Study

Programming Languages

- HTML
- CSS
- JavaScript

Development Environment

- Visual Studio Code

Browser Testing

- Google Chrome

5. Problem Statement / Case Study Implementation Details and Snapshots.

The project is implemented using **HTML, CSS, and JavaScript files**.
Each file handles a specific part of the application such as UI design, game logic, leaderboard system, and forum interaction.

Project Files

HTML Files

- **index.html** – Landing page and navigation of the application

- **flappybird.html** – Main gameplay interface
- **leaderboard.html** – Displays player rankings
- **forum.html** – Community comment page

CSS Files

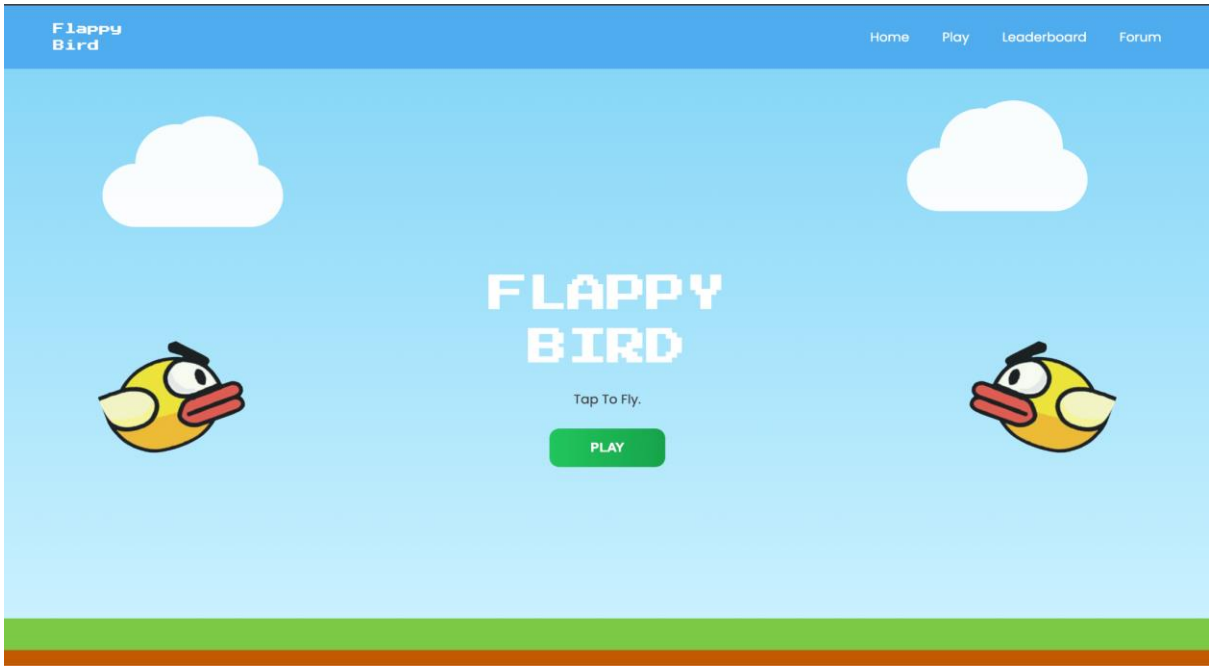
- **style.css** – Styling for the home page
- **flappybird.css** – Styling for the gameplay screen
- **leaderboard.css** – Styling for the leaderboard layout
- **forum.css** – Styling for the forum page

JavaScript Files

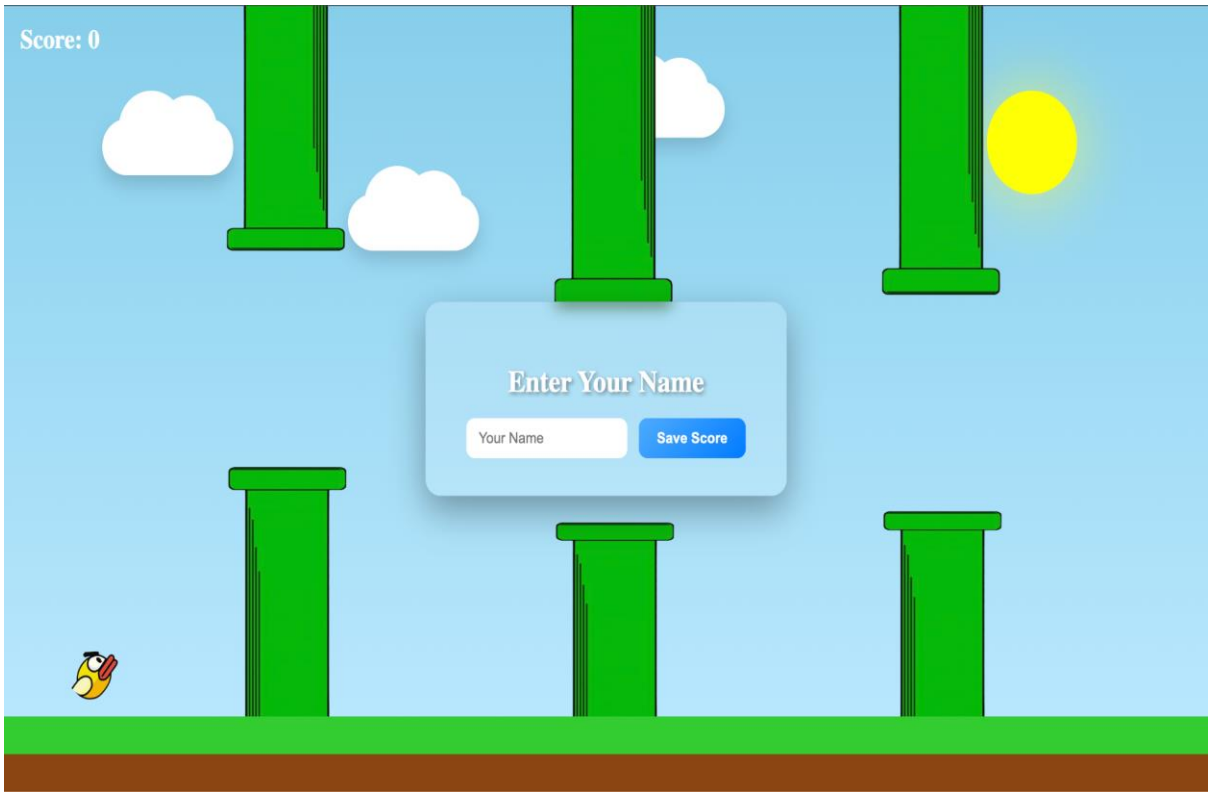
- **flappybird.js** – Game physics, pipe generation, collision detection, and score logic
- **leaderboard.js** – Retrieves scores from localStorage and displays player rankings
- **forum.js** – Handles comment posting and storage using localStorage

Assets Folder

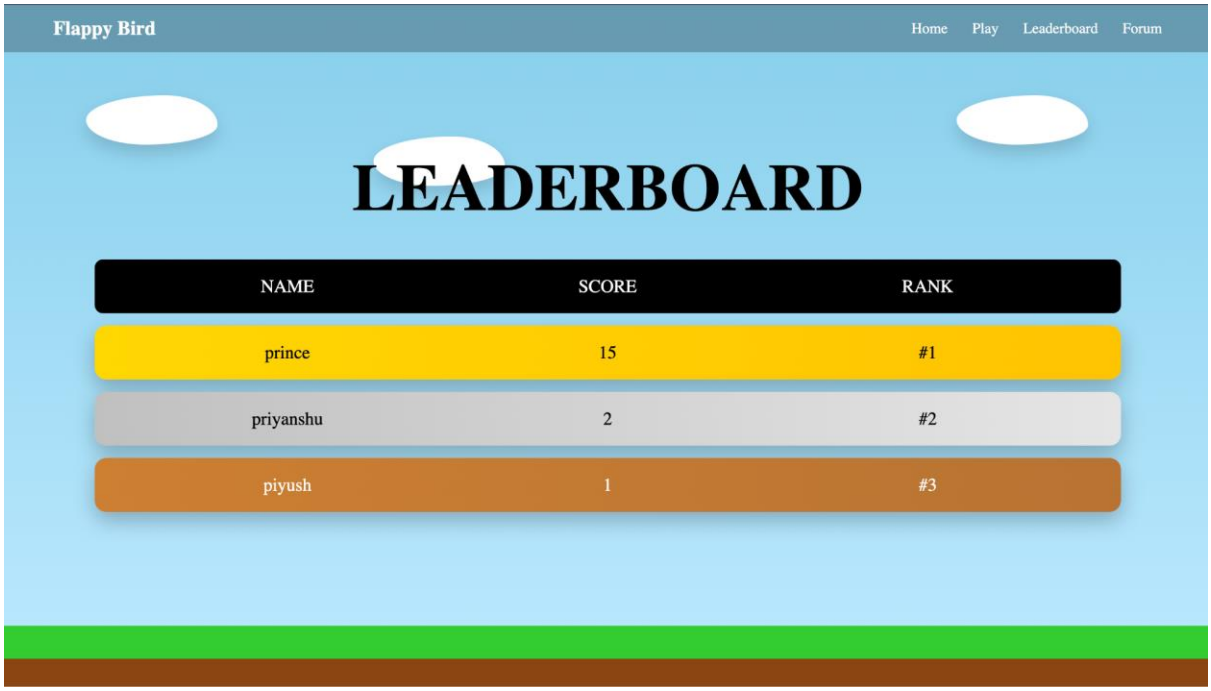
- **assets/** – Contains images used in the game such as bird and pipe graphics.



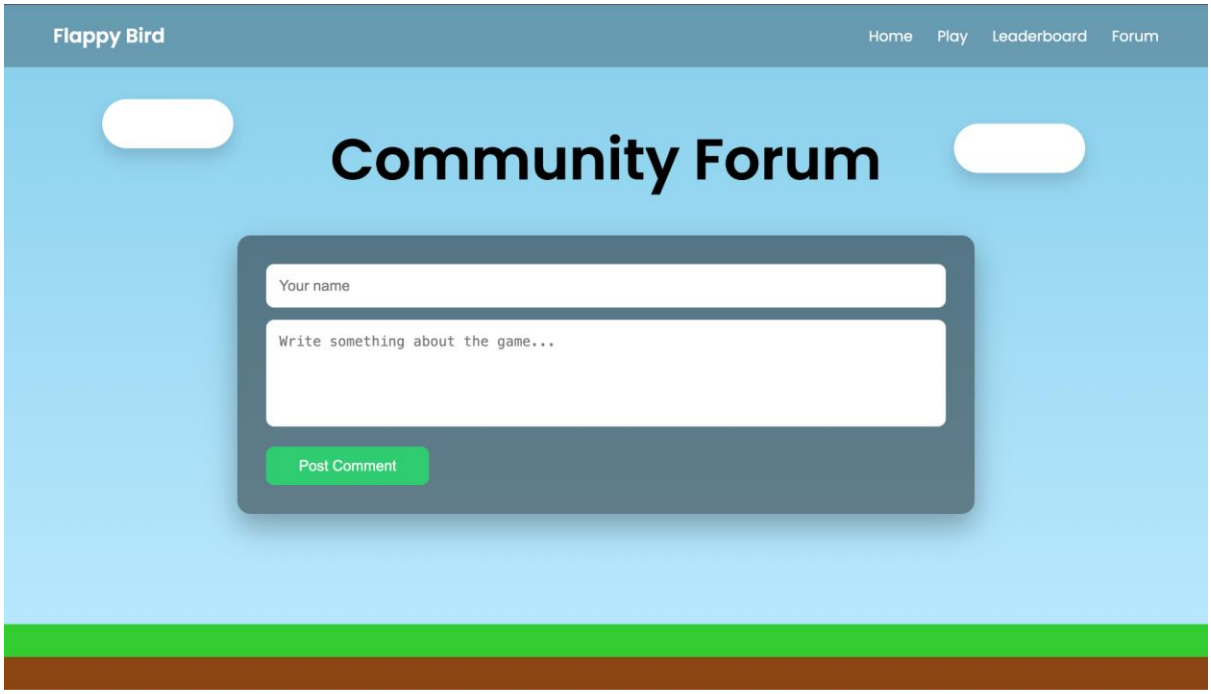
Above image is of **Home page**



Above image is of **Gamescreen** after the game is over



Above image is of **Leaderboard page**



Above image is of **Forum page**

6. Problem Statement / Case Study Results and Conclusion.

Findings

- The implemented game successfully demonstrates basic **game physics and interaction using JavaScript**.
- Using **localStorage** allows player scores and forum comments to persist without a backend server.
- The leaderboard system effectively ranks players and updates existing scores to avoid duplicate entries.
- The forum feature improves user engagement by allowing players to share feedback.

Conclusion

This case study demonstrates how a browser-based arcade game can be developed using HTML, CSS, and JavaScript. The system integrates gameplay mechanics, score tracking, leaderboard ranking, and a forum for user interaction.

The project provides a practical example of front-end web development techniques applied to interactive game design. It also highlights how persistent data storage using localStorage can be used to maintain user information without requiring a database or backend server.

7. References

- MDN Web Docs – JavaScript Documentation
- W3Schools – HTML, CSS, and JavaScript Tutorials
- Stack Overflow – Web Development Discussions
- Flappy Bird Game Design References